SIT 210 TASK 3.1P

1. GitHub Repository link
   https://github.com/HollowOWisp/SIT210-3.1P
2. Block Diagram of the System

   Hardware & Software used

   Arduino Nano 33 IoT
   PIR Motion Sensor
   Breadboard and jumper wires
   Arduino IDE
   Node-RED
   HiveMQ Cloud (MQTT Broker)
   Gmail (Email notification service & app password generator)

3. Trigger Mechanism

   The trigger mechanism is incorporated in the Arduino program. The motion sensor always keeps track of the environment and returns a HIGH value if motion is detected.

The Arduino code then reads this value and sends a message to the MQTT topic:

"Motion detected" if motion is detected
"No motion detected" if no motion is detected

```cpp


#define MQTT_SERVER "d83b06ae45974bdea2e0adfb8248d669.s1.eu.hivemq.cloud"
#define MQTT_PORT 8883
#define MQTT_TOPIC "sensor/motion"
#define MQTT_USER "YOUR_MQTT_USERNAME"
#define MQTT_PASSWORD "YOUR_MQTT_PASSWORD"


const int pirSensorPin = 2;
int motionDetected = 0;
int previousMotionState = LOW;


WiFiSSLClient wifiSSLClient;
PubSubClient mqttClient(wifiSSLClient);

void connectMQTT() {
  while (!mqttClient.connected()) {
    Serial.println("Connecting to MQTT...");
    if (mqttClient.connect("ArduinoNanoIoT", MQTT_USER, MQTT_PASSWORD)) {
      Serial.println("Connected to MQTT Broker!");
    } else {
      Serial.print("Failed, rc=");
      Serial.print(mqttClient.state());
      Serial.println(" Retrying in 5 seconds...");
      delay(5000);
    }
  }
}

void setup() {
  Serial.begin(115200);
  pinMode(pirSensorPin, INPUT);

  WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
  while (WiFi.status() != WL_CONNECTED) {
    delay(1000);
    Serial.println("Connecting to WiFi...");
  }
  Serial.println("Connected to WiFi");
```

Output

```
Sketch uses 21216 bytes (8%) of program storage space. Maximum is 262144 bytes.
Global variables use 3952 bytes (12%) of dynamic memory, leaving 28816 bytes for local variables. Maximum is 32768 bytes.
```

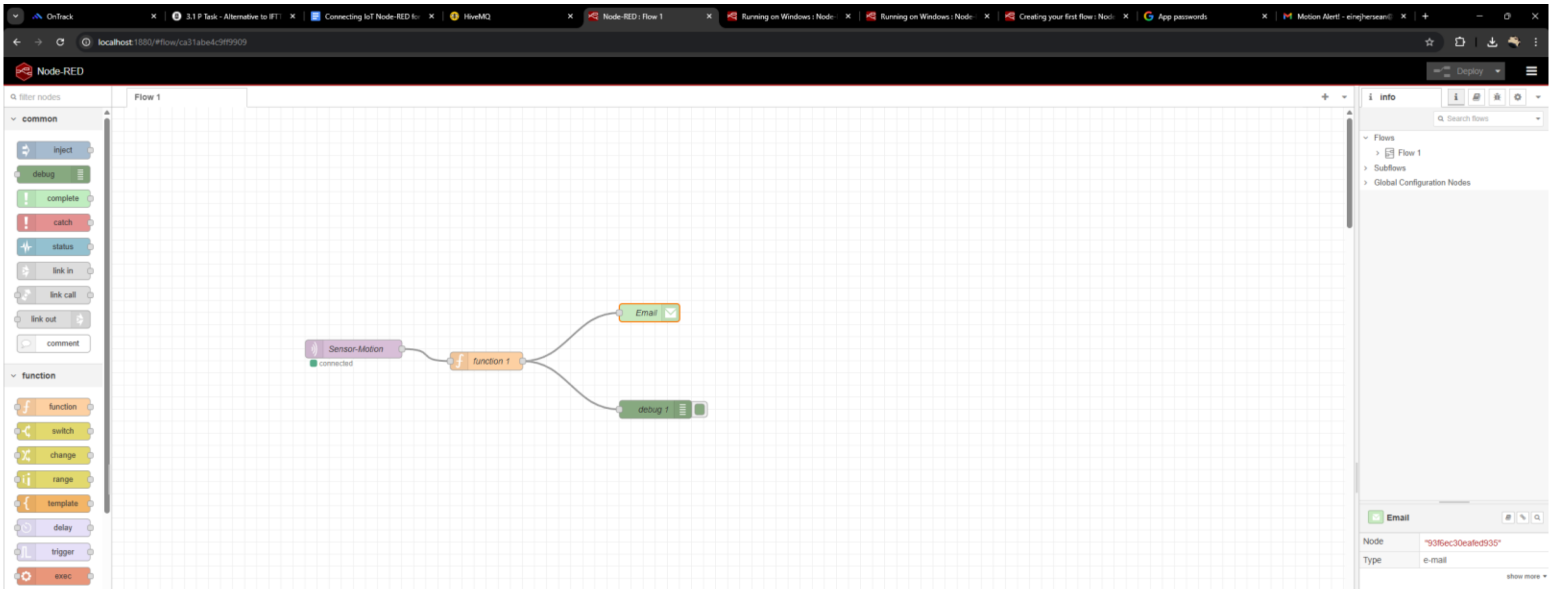
This is done by reading the sensor value using digitalRead(). Depending on the sensor value, a message is sent to the MQTT server using mqttClient.publish().

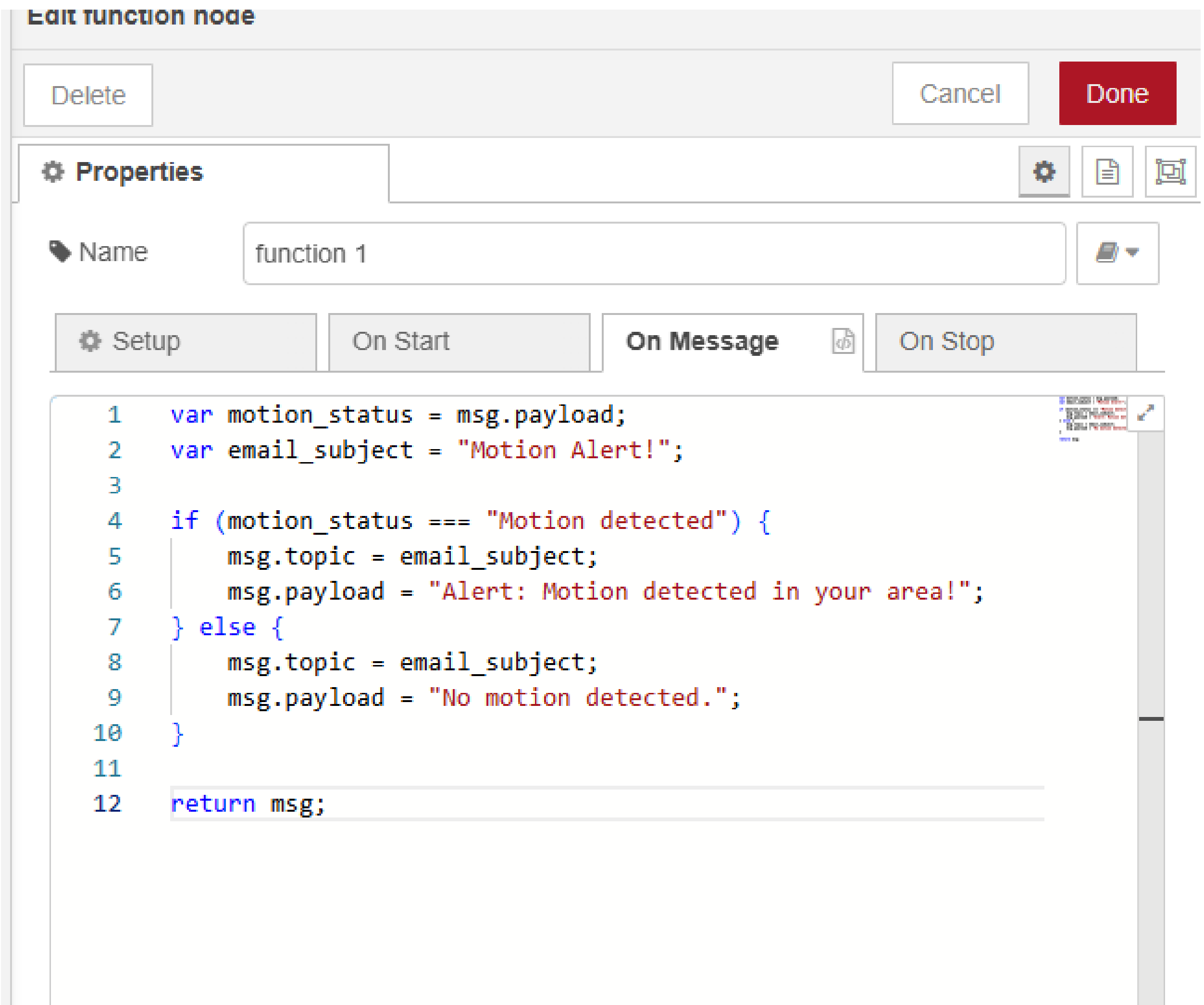
3a. Notification Mechanism

The notification mechanism is facilitated using Node-RED.

Node-RED subscribes to the MQTT topic, which is sensor/motion, to receive messages from the Arduino board using the MQTT protocol. It receives the messages sent by the Arduino board.

The node processes the message, which is then formatted to make it a notification email using the email node, which sends the email using the Gmail SMTP server.


Node-RED
Deploy
Flow 1
common
inject
debug
complete
catch
status
link in
link call
link out
comment
function
function
switch
change
range
template
delay
trigger
exec
Sensor-Motion
connected
function 1
Email
debug 1
info
Flows
Flow 1
Subflows
Global Configuration Nodes
Email
Node
"93f6ec30eafed935"
Type
e-mail
show more

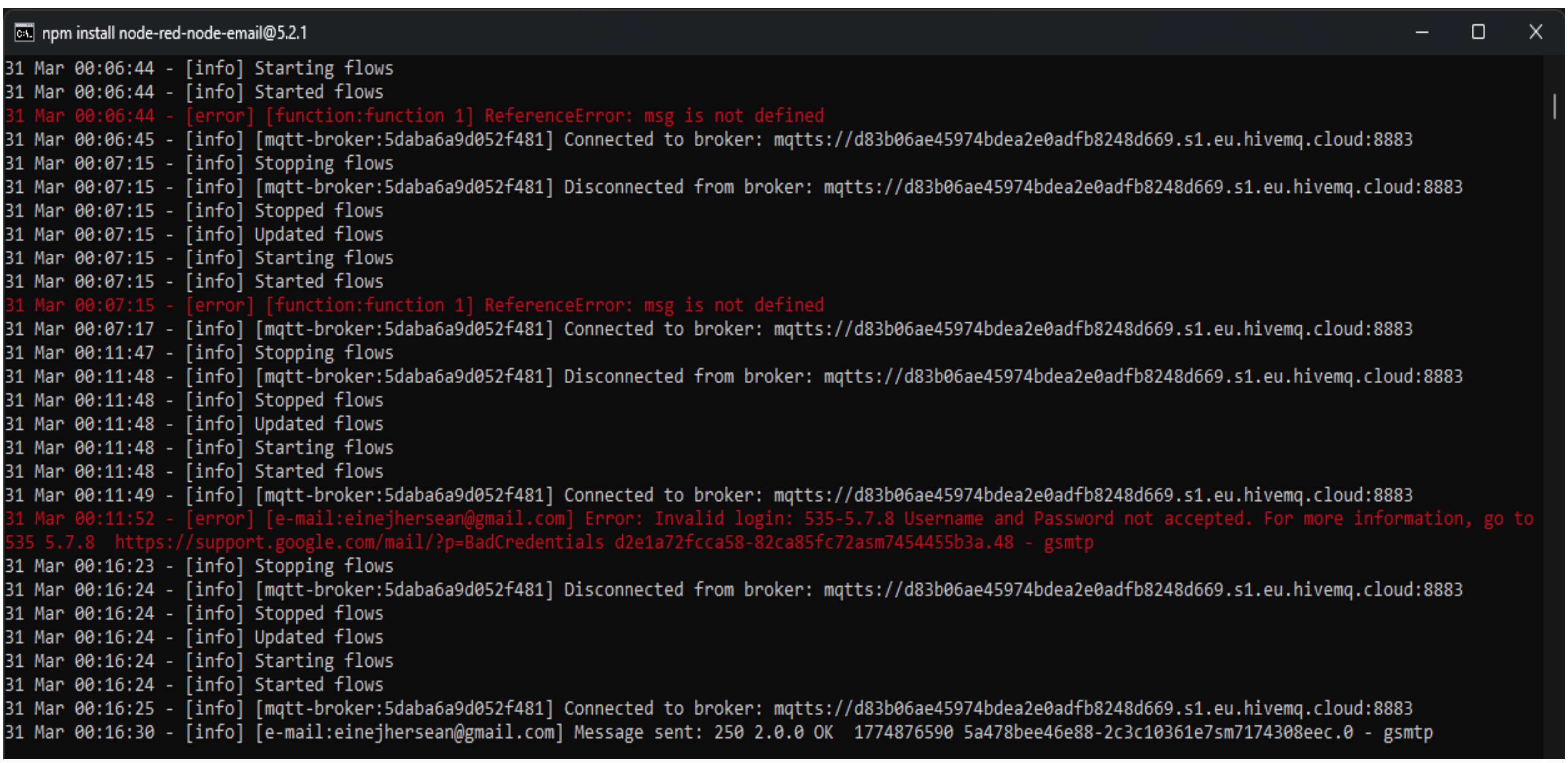

Edit function node
Delete
Cancel
Done
Properties
Name
function 1
Setup
On Start
On Message
On Stop
var motion_status = msg.payload;
var email_subject = "Motion Alert!";
if (motion_status === "Motion detected") {
msg.topic = email_subject;
msg.payload = "Alert: Motion detected in your area!";
} else {
msg.topic = email_subject;
msg.payload = "No motion detected.";
}
return msg;

The MQTT In node receives the message from the broker, The Function node checks if the motion is detected, and the Email node sends the alert to the user. This way, the user receives real-time alerts when the motion is detected or stopped.

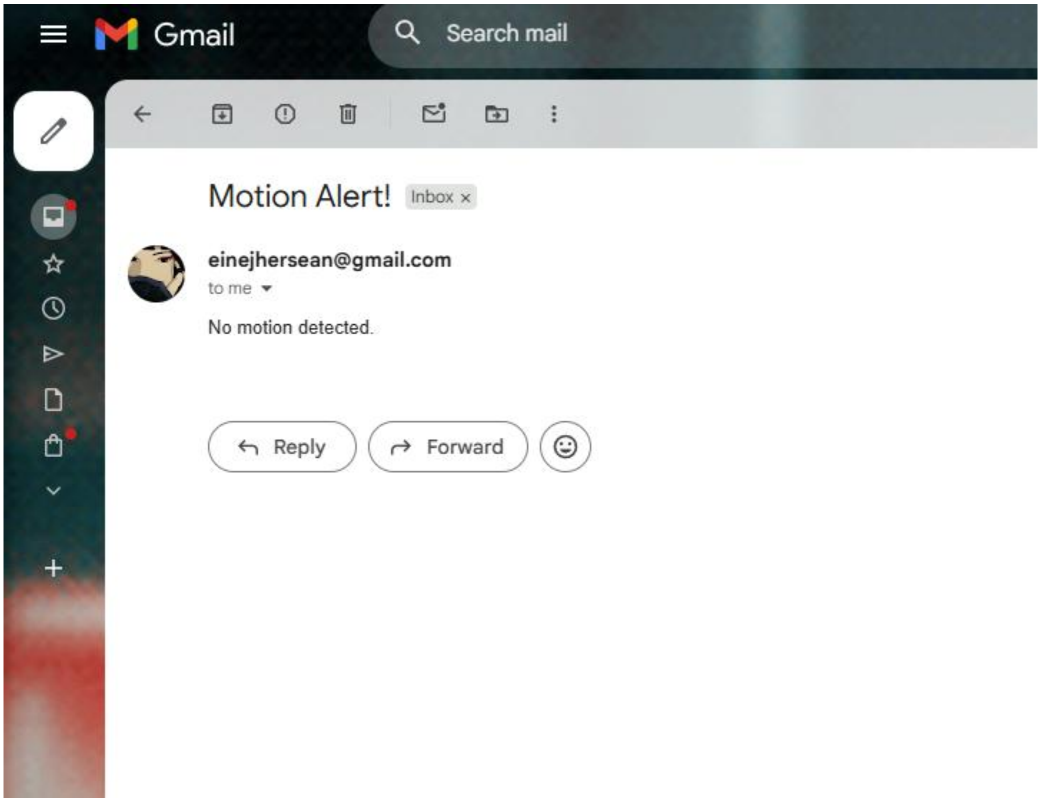
4. Testing the System

Through the sensor's response and verification of the transmission of the message, it was tested. The Arduino system was able to sense that something had changed, like if a person walked in front of it, and sent out a message through MQTT. Another type of message was sent if there was no motion.

Node-RED was able to receive these messages and send out email alerts as responses. For verification that messages were sent and that they accurately represented the state of the sensor, tests were conducted.



```
npm install node-red-node-email@5.2.1
31 Mar 00:06:44 - [info] Starting flows
31 Mar 00:06:44 - [info] Started flows
31 Mar 00:06:44 - [error] [function:function 1] ReferenceError: msg is not defined
31 Mar 00:06:45 - [info] [mqtt-broker:5daba6a9d052f481] Connected to broker: mqtts://d83b06ae45974bdea2e0adfb8248d669.s1.eu.hivemq.cloud:8883
31 Mar 00:07:15 - [info] Stopping flows
31 Mar 00:07:15 - [info] [mqtt-broker:5daba6a9d052f481] Disconnected from broker: mqtts://d83b06ae45974bdea2e0adfb8248d669.s1.eu.hivemq.cloud:8883
31 Mar 00:07:15 - [info] Stopped flows
31 Mar 00:07:15 - [info] Updated flows
31 Mar 00:07:15 - [info] Starting flows
31 Mar 00:07:15 - [info] Started flows
31 Mar 00:07:15 - [error] [function:function 1] ReferenceError: msg is not defined
31 Mar 00:07:17 - [info] [mqtt-broker:5daba6a9d052f481] Connected to broker: mqtts://d83b06ae45974bdea2e0adfb8248d669.s1.eu.hivemq.cloud:8883
31 Mar 00:11:47 - [info] Stopping flows
31 Mar 00:11:48 - [info] [mqtt-broker:5daba6a9d052f481] Disconnected from broker: mqtts://d83b06ae45974bdea2e0adfb8248d669.s1.eu.hivemq.cloud:8883
31 Mar 00:11:48 - [info] Stopped flows
31 Mar 00:11:48 - [info] Updated flows
31 Mar 00:11:48 - [info] Starting flows
31 Mar 00:11:48 - [info] Started flows
31 Mar 00:11:49 - [info] [mqtt-broker:5daba6a9d052f481] Connected to broker: mqtts://d83b06ae45974bdea2e0adfb8248d669.s1.eu.hivemq.cloud:8883
31 Mar 00:11:52 - [error] [e-mail:einejhersean@gmail.com] Error: Invalid login: 535-5.7.8 Username and Password not accepted. For more information, go to
535 5.7.8  https://support.google.com/mail/?p=BadCredentials d2e1a72fcca58-82ca85fc72asm7454455b3a.48 - gsmtp
31 Mar 00:16:23 - [info] Stopping flows
31 Mar 00:16:24 - [info] [mqtt-broker:5daba6a9d052f481] Disconnected from broker: mqtts://d83b06ae45974bdea2e0adfb8248d669.s1.eu.hivemq.cloud:8883
31 Mar 00:16:24 - [info] Stopped flows
31 Mar 00:16:24 - [info] Updated flows
31 Mar 00:16:24 - [info] Starting flows
31 Mar 00:16:24 - [info] Started flows
31 Mar 00:16:25 - [info] [mqtt-broker:5daba6a9d052f481] Connected to broker: mqtts://d83b06ae45974bdea2e0adfb8248d669.s1.eu.hivemq.cloud:8883
31 Mar 00:16:30 - [info] [e-mail:einejhersean@gmail.com] Message sent: 250 2.0.0 OK  1774876590 5a478bee46e88-2c3c10361e7sm7174308eec.0 - gsmtp
```


Gmail
Search mail
Motion Alert!
Inbox ×
einejhersean@gmail.com
to me
No motion detected.
Reply
Forward